

DevOps lab Program 5: Demonstrate CI Integration by Running Jenkins in a Docker Container

Dockerfile

```
1RV24MC075_Pragathi_h_r@dell: ~/prg5
GNU nano 6.2 Dockerfile
FROM jenkins/jenkins:lts
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

docker-compose.yml

```
1RV24MC075_Pragathi_h_r@dell: ~/prg5
GNU nano 6.2 docker-compose.yml
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home
```

Build and run Jenkins:

```
1RV24MC075_Pragathi_h_r@dell:~$ cd prg5
1RV24MC075_Pragathi_h_r@dell:~/prg5$ nano Dockerfile
1RV24MC075_Pragathi_h_r@dell:~/prg5$ nano docker-compose.yml
1RV24MC075_Pragathi_h_r@dell:~/prg5$ nano Dockerfile
1RV24MC075_Pragathi_h_r@dell:~/prg5$ docker build -t prg5 .
[+] Building 80.0s (6/6) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default
=> => transferring dockerfile: 136B                                              0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts                  79.9s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 2B                                                  0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:7b1c378278279c8688efd6168c25a1c2723a6bd6f0420beb5ccefabee3cc3bb1 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io             0.0s
=> => exporting to image                                                         0.0s
=> => exporting layers                                                         0.0s
=> => writing image sha256:8568f33a5faf5509365f35e7ba295e71f8e9c7cfe269f5657f681cf5e0fcf217 0.0s
=> => naming to docker.io/library/prg5                                         0.0s
```

```
1RV24MC075_Pragathi_h_r@dell: ~/prg5
1RV24MC075_Pragathi_h_r@dell:~/prg5$ docker run -it -p 8080:8080 fifth
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-11-19 10:32:42.423+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2025-11-19 10:32:43.228+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-11-19 10:32:43.297+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.21
9Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea; jvm 21.0.9+10-LTS
2025-11-19 10:32:43.670+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find
d.org.eclipse.jetty.ee9.jsp.JettyJspServlet
2025-11-19 10:32:43.730+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2025-11-19 10:32:44.098+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found
at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2025-11-19 10:32:44.226+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHand
ler@3c782d8e{Jenkins v2.528.2,/,b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHa
ndler@572e6fd9{STARTED}}
2025-11-19 10:32:44.237+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bca6fac{HTTP/1.1, (h
ttp/1.1)}{0.0.0.0:8080}
2025-11-19 10:32:44.249+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@3668d4{STARTING}[12.0.25
,sto=0] @2358ms
2025-11-19 10:32:44.250+0000 [id=40] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2025-11-19 10:32:44.378+0000 [id=39] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.2
2025-11-19 10:32:44.467+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2025-11-19 10:32:44.478+0000 [id=59] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2025-11-19 10:32:45.211+0000 [id=47] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2025-11-19 10:32:45.215+0000 [id=47] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2025-11-19 10:32:45.215+0000 [id=47] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2025-11-19 10:32:45.364+0000 [id=52] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2025-11-19 10:32:45.364+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2025-11-19 10:32:45.365+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2025-11-19 10:32:45.366+0000 [id=50] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2025-11-19 10:32:45.383+0000 [id=75] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2025-11-19 10:32:45.701+0000 [id=60] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> b3fcfe8bd9834e629ff4de53b7dcff4e
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
```

Browser Navigation:



